INCH-POUND

A-A-59795A
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, CONTAINER HANDLER, DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, ROUGH TERAIN 50,000-POUND CAPACITY

The General Services Administration has authorized the use of this Commercial Item Description (CID) as a replacement for MIL-T-52951 for all federal agencies.

1. SCOPE. This CID covers the general requirements for a commercial, diesel-engine-driven, all-wheel-drive, pneumatic tired, rough terrain container handling forklift truck (RTCH). The RTCH will primarily be used to lift, transport, and stack International Standards Organization (ISO) containers and sea/land containers of 20 and 40 foot lengths on paved, unimproved, mud, and snow surfaces. When using forks in the bottom lift configuration, the vehicle will have the capability to lift, transport, and stack loads up to 50,000 pounds at a 48-inch load center. When using a top handling attachment, the vehicle will have the capability to lift, transport, and stack loads (not including attachment) up to 44,000 pounds at a 48-inch load center.

2. SALIENT CHARACTERISTICS

2.1 Safety. The forklift shall comply with ASME/ANSI B56.6, UL 558, and OSHA standards in effect at the time of manufacture. An operator's seatbelt conforming to SAE J386, and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tipover shall be provided.

2.2 Engine. The forklift shall be powered by a standard, commercial, liquid cooled, diesel engine. The engine shall operate on diesel fuel in accordance with ASTM D975 without detrimental effects on the engine, or its performance.

2.3 Fuel system. The forklift shall have all necessary filters, water separators, and components required for diesel fuel operation. The forklift shall have a fuel tank of sufficient capacity to allow at least eight hours of continuous operation.

Beneficial comments, recommendations, additions, deletions, clarifications, and any data which may improve this document should be sent to:
U.S. AIR FORCE, AFLCMC/WN-SEV, 295 Byron Street, Robins AFB, GA 31098 or emailed to SPEC99@us.af.mil/ .

AMSC N/A FSC 3930

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2.4 Starting system. The starter switch shall not activate the engine starter while the engine is running, nor when the engine is not running and the transmission is in any forward or reverse gear. The starter switch shall only operate when the transmission is in the "neutral" or "park" positions.

2.5 Transmission. A torque converter and a powershift type of transmission shall be furnished, with at least three speeds forward and three speeds reverse. The transmission shall provide for positive inching control, or declutch control, that permits lifting of rated load at maximum lifting speed while the transmission control is in either forward or reverse direction, without vehicle movement. This control shall also apply the service brakes. When specified (see 6.2c), a transmission oil cooler shall be furnished.

2.6 Hydraulic system. A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system. The hydraulic system shall also be designed to prevent forward tilt in the event of system failure; tilt angle shall not deviate more than two degrees if failure occurs at any position during load handling operations. The hydraulic lift system shall be designed so that in the event of hydraulic hose or pump failure, the load shall not descend (freefall) at an uncontrolled rate; lowering speed shall not exceed eight (8) inches per second under any load condition. A manual override for the tilt function shall also be provided to allow for load removal.

2.7 Uprights and carriage. With no load on the forks, the seated operator with seatbelt engaged shall be able to see at least one fork tip at all lift heights and fork positions.

2.8 Forks and carriage. Fork tines shall have the following dimensions. The forks shall be mounted so that when fully lowered and forklift operating in reverse, there is no damage or degradation to the fork tines, their attachment, or any other component of the carriage, including fork positioner and side shift.

a. Length: 96 inches (maximum)

b. Width: 11.8 inches (maximum)

c. Thickness: 4.5 inches maximum, except bend in fork heel may exceed 4.5 inches

d. Fork tip thickness: 1 ¼ inch (maximum)

e. Fork taper (underside of forks): Taper from fork tip rearward shall be 30 inches (minimum)

f. Fork carrier assembly width shall not exceed the overall width of the forklift

g. Fork tines shall be 48 inches in length with an underside taper. Fork tip thickness shall not exceed 1 ¼ inches

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2.9 Fork positioner. A hydraulically powered fork positioner shall be furnished and controlled by the operator while seated.

2.10 Side shift. Hydraulically powered side shift capability shall be furnished. It shall be controlled by the seated operator, and shall provide at least 12-inch side shift capability (each side of center or 24-inches total travel).

2.11 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical rear load guard of at least 102.5 inches high, measured from the load carrying surfaces of the forks.

2.12 Steering. Power steering, with emergency steering in the event of power failure, shall be furnished.

2.13 Service brakes. Power brakes conforming to ASME/ANSI B56.6 shall be furnished, except maximum brake pedal force during stopping shall not exceed 80 pounds.

2.14 Parking brake. Parking brake shall conform to ASME/ANSI B56.6

2.15 Electrical system. A 24 volt direct current (VDC) electrical system shall be furnished. The electrical system shall consist of all electrical components necessary for operation of the forklift. The forklift shall have sufficient electrical grounding to prevent static discharge.

2.15.1 Backup alarm. An audible backup alarm shall be furnished. The alarm shall automatically activate when the transmission selector is placed in reverse.

2.16 Instruments and controls. In addition to the instruments supplied on the standard commercial forklift, an hour meter shall be installed in the instrument panel. Except the hour meter, all instruments shall illuminate. All load motion controls shall be right hand operation controls and of the self-centering type, meaning; the controls shall return to the neutral position when released.

2.17 Lighting. Minimum of eight (8) sealed beam floodlights, six (6) in front facing forward, two of which shall illuminate the two near twist locks for all lift heights or shall be adjustable from the operators position, and two mounted in the rear facing rearward, shall be furnished. An additional floodlight shall be furnished that is adjustable from inside the cab by the operator while seated; the floodlight shall provide at least 270° rotation capability. Tail light(s) and brake stop light(s) shall be installed. All lights shall be protected by location or guards. Individual operator controlled switches shall be provided for front floodlights, rear floodlights, and adjustable lights.

2.18 Horn. A horn shall be furnished, with the horn actuator located within reach of the seated operator.

2.19 Operator's overhead guard. The guard shall be in accordance with ASME/ANSI B56.6. It shall also be a falling-object-protective-structure (FOPS) conforming to ASME/ANSI B56.6.

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2.20 Cab. The forklift shall be equipped with an enclosed cab that shall incorporate overhead protection requirements of ASME/ANSI B56.6. The cab shall have a door on each side of the vehicle with at least one window per door which can be opened for ventilation; the windows shall be capable of being locked in either the open or closed positions. The doors shall be provided with hinges that can be serviced without door removal and shall allow adjustments for door fit. The doors shall be capable of being locked in either the opened or closed positions. In addition to the door windows, the cab shall be furnished with a front windshield, rear window, and side windows. The windshield and all windows shall be safety glass. The cab interior shall be furnished with water-resistant non-absorbent thermal insulation. The cab shall be equipped with air conditioning and an overhead fan. A commercial fire extinguisher shall be mounted on the truck in an easily accessible location. The fire extinguisher shall be a minimum 2-1/2 pound capacity ABC type, or equivalent. The fire extinguisher shall be UL listed.

2.21 Windshield wipers. There shall be at least two wipers furnished, one on the front windshield, and one on the rear window.

2.22 Winterization kits. There shall be Type A (-65°F) and Type C (-25°F) options available for winter protection.

2.22.1 Type A winterization. When specified (see 6.2b), the truck shall be furnished with Type A winterization that shall protect to -65° F and shall consist of the following:

a. Power plant heaters. Engine coolant, engine oil and battery heaters shall be provided. All heaters shall operate on 110 volt, alternating current. A three wire, 25 feet long weatherproof cable of adequate capacity for all heaters being used simultaneously shall be provided. A stowage place shall be provided on the truck to keep the cable when not in use. The heaters shall be as follows:

(1) The coolant heater shall be installed in the engine block or lower coolant inlet hose. A coolant-circulating pump, driven by a 110-volt alternating current motor shall be provided when a coolant inlet hose heater is furnished. The heater shall have adequate capacity to maintain engine coolant at a temperature of 10° F in an ambient temperature of -65° F. It shall be controlled to limit engine coolant to not more than 150° F.

(2) An engine oil heater with adequate capacity to maintain engine oil at a temperature of at least +10 degrees F in an ambient temperature of -65° F shall be furnished. It shall be controlled to limit engine oil temperature to not more than 150 ° F.

(3) A battery heater shall be provided. It shall have adequate capacity to maintain battery electrolyte at a temperature of at least +10° F in an ambient temperature of -65°F, and shall have a thermostat to limit temperature of the electrolyte to not more than +80°F.

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b. Cab. The cab shall meet the requirements of 3.20, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 3.23.1.c.

c. Heater and defroster. The heater shall be of sufficient capacity to maintain a temperature of +40° F at cab floor level in an ambient temperature of -40° F.

d. Cold starting aid. A measured shot ether system or glow plug(s) shall be provided to assist in engine cold temperature starting.

e. Antifreeze. Engine coolant shall be protected to -65° F.

2.22.2 Type C winterization. When specified (see 6.2(b), the truck shall be furnished with Type C winterization that shall protect to -25° F and shall consist of the following:

a. Cold starting aid. Glow plug(s) or a measure shot ether injection system shall be furnished.

b. Cab. The cab shall meet the requirements of 3.20, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 3.23.2.c.

c. Cab heater and defroster. The heater shall have sufficient capacity to maintain a temperature of at least +40° F at cab floor level in an ambient temperature of -20° F.

d. Antifreeze. Engine coolant shall be protected to -25° F with antifreeze.

2.23 Performance.

a. Capacity. Capability to safely handle a 50,000-pound load with a 48-inch load center, at all lift heights.

b. Lifting speed. At least 30 fpm.

c. Lowering speed. Not more than 80 fpm with rated load on forks and not more than 35 fpm unloaded.

d. Collapsed mast height. Upright height with forks on ground (collapsed mast height): 228 inches maximum, with no load on the forks.

e. Maximum forklift height. At least 143 inches with rated load on forks and forks horizontal 113.50 inches (with 20’ or 40’ handler kit).

f. Right angle turn. With rated load on the forks, forklift positioned perpendicular to a wall, and with the front of the load against the wall, the truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 210 inches.

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g. Travel speed. 16 miles per hour (mph) minimum in both forward and reverse, while carrying rated load on the forks. The vehicle speed shall be governed so that maximum speed for safe operations is not exceeded.

h. Slope ascension. With the truck facing up the slope and with rated load on the forks, the truck shall be able to accelerate up a 25 percent slope from a dead stop.

i. Upright tilt. With no load on the truck, the forks shall have at least 5 of forward tilt, and at least 8 degrees of rear tilt.

j. Ground clearance. With rated load on the forks and mast vertical, ground clearance beneath the mast shall be at least 12 inches. With no load on the forks, the ground clearance of the truck other than the mast and axles shall be at least 7.0 inches.

k. Load drift. With hydraulic fluid at normal operating temperature, the lift assembly shall hold rated load at maximum lift height for at least 2.0 minutes with not more than 1.0 inch of vertical drift, and not more than 0.5 degree of rotational drift.

l. Stability. The forklift shall meet the ASME/ANSI B56.1 "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements.

m. Noise limits. Maximum allowable noise level shall not exceed 84dB at the operator's ear. If the noise level at the operator's ear exceeds 84 dB (85 to 92 dB) a permanent warning decal or placard shall be installed inside the cab clearly visible to the operator. Noise level limits that exceed 92dB are not acceptable.

2.24 Painting. The forklift shall be primed and painted using standard commercial practices. Unless otherwise specified (see 6.2d). When specified the manufacturer must provide Forest Green (24052 of SAE AMS-STD-595) or Desert Sand (30313 of SAE AMS-STD-595).

2.25 Markings, data plates, and instruction plates. Corrosion resistant identification plate, and all warning and instruction plates shall be provided and mounted on the vehicle. The identification plate shall contain the following information:

- Truck, Lift, Fork
- Capacity: 50,000 pounds at (48) inch load center
- Gross Vehicle Weight (GVW):
- National Stock Number (NSN):
- Contract Number:
- Serial Number:
- Model Number:
- Registration Number:
- Manufactured by:
- Name:
- Address:
- Service Phone Number:

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- US Property:

2.25 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip-coating compound, or be furnished with a nonslip metal or treadplate surface.

2.26 Slinging and tie down provisions. Permanently affixed slinging and tie down provisions that enable the truck to be lifted completely assembled and in its normal operational mode, and to be fastened to the floor or deck of a transportation medium, shall be provided. Provisions shall conform to MIL-STD-209, Classes1 and 2 or Class 3 for Type II equipment.

2.27 Electromagnetic interference (EMI). The forklift shall comply with EMI requirements of SAE J551-1 and SAE J551-2.

2.28 Tropical rust proofing. When specified (see 6.2e), tropical proofing shall be provided. The forklift shall be treated with a corrosion preventative compound in accordance with A-A-59295. At a minimum, these areas shall be coated:

a. Cab, interior

b. Cab, exterior

c. Seams

d. Welds

e. Hidden recessed areas

f. Location where dissimilar metals come in contact

g. Area above fuel tank(s)

h. Engine oil pan and transmission oil pan

i. Radiator support structure

j. Battery box

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. The contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the forklift.

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3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The RTCH will be in compliance with ISO 9001. The government reserves the right to require proof of such conformance.

5. PACKAGING

5.1 Preservation, Packaging, Labeling and Marking. Unless otherwise specified, the preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions and handling. This involves shipment from the supply source to the receiving activity, and reshipment from the receiving activity. The preservation, packaging, and packing shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling and marking shall be as specified in the contract and/or order.

6 NOTES

6.1 Source of documents.

6.1.1 Government documents. Copies of these documents are available online at https://quicksearch.dla.mil/ or from the DLA Document Services, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.1.2 ASME /ANSI documents may be obtained at http://www.asme.org/ or from American Society of Mechanical Engineers, P.O. Box 2900 Fairfield NJ 07007.

6.1.3 SAE documents may be obtained at http://www.sae.org/ or from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale PA 15096.

6.1.4 Underwriters Laboratories (UL) UL documents may be obtained at http://ulstandardsinfonet.ul.com/catalog or hardcopies and CD's can be found at COMM 2000 1414 Brook Drive Downers Grove, IL 60515

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6.1.5 Copies for Occupational Safety and Health Association (OSHA) may be obtained online at http://www.osha.gov/ or be sent to the U. S. Department of Labor, 200 Avenue NW, Room 423, Washington DC 20210

6.1.6 ISO standards may be obtained at http://www.iso.org/iso/en/ISOOnline.frontpage or hardcopies may be found at International Organization for Standardization (ISO) 1, rue de Varembé, Case postale 56 CH-1211 Geneva 20, Switzerland

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, CID number, and date of this document

b. Winterization (type A or Type C) if required (see 2.22)

c. Transmission oil cooler, if required (see 2.5)

d. Fork Lift primed and painted, if different (see 2.24)

e. Tropical Rust Proofing, if required (see. 2.28)

f. Packaging requirements, if different (see 5.1).

6.3 Subject term (key word) listing.

All-wheel-drive
ISO transporter
Lift capacity, 50,000-pounds
RTCH (Rough Terrain Container Handler)

6.4 Changes from previous issue. The margins of this CID are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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